D(n)-tuples in a non-commutative ring

Problèmes ouverts — A. Dujella · 2026

Problem 1.10 (Dujella, 2026). Define an analogue of a $D(n)$-m-tuple in a non-commutative ring and prove some non-trivial results concerning the existence of $D(n)$-quadruples, e.g. in the ring of 2×2 matrices with integer entries.

Remark. Dujella and Franušić (arXiv :2604.23404) proposed considering the Jordan product $A \circ B = \frac{1}{2}(AB + BA)$. The notion of regular triples naturally generalizes in this situation : if $A \circ B = R^2$, then $C = A + B + 2R$ satisfies $A \circ C = (A + R)^2$ and $B \circ C = (B + R)^2$.